

Section: Module 4

Linear Regression and Randomized Experiments

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Logistics

- Midterm: October 14th, 3 hours
 - Review Section: October 7th
- What you should do before midterm
 - Solve review questions
 - Solve practice midterm
 - Check solutions of problem sets
 - Understand *details* of class materials
- Important to be able to follow the logic
 - This time, section materials can be a bit overwhelming
 - Try to follow each line carefully

Today's Agenda

- Review of Linear Regression
 - Basic Properties
 - Best linear approximation
 - Ordinal Least Squares (OLS)
 - Frisch-Waugh-Lowell (FWL) theorem
 - Asymptotic Variance (Extra slides, optional)
- Regression and Causal Inference
 - Case 1: No Control
 - Simple regression review questions: OLS variance and its bias
 - Case 2: With control (no interaction)
 - Case 3: With control (with interaction)
 - Recommended reading: Ch7 of Imbens and Rubin (2015)
 - Case 4: Stratified Design
- Cluster Randomized Experiment
 - Related to Problem Set 4

Review: Linear Regression (1)

- Linear regression with K predictors is written as

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_K X_{Ki} + \epsilon_i, \quad \underbrace{\mathbb{E}[\epsilon_i \mid X_{1i}, \dots, X_{Ki}] = 0}_{\text{Strict Exogeneity}}$$

- Let $\mathbf{X}_i = [1, X_{1i}, \dots, X_{Ki}]^\top$ and $\boldsymbol{\beta} = [\beta_0, \dots, \beta_K]^\top$. Then,

$$Y_i = \mathbf{X}_i^\top \boldsymbol{\beta} + \epsilon, \quad \mathbb{E}[\epsilon_i \mid \mathbf{X}_i] = 0$$

- Equivalently,

$$\mathbb{E}[Y_i \mid \mathbf{X}_i] = \mathbf{X}_i^\top \boldsymbol{\beta}$$

- Regression is conditional expectation function of Y_i given $\mathbf{X}_i$

Review: Linear Regression (2)

- Note that $\mathbb{E}[\epsilon_i | \mathbf{X}_i] = 0$ implies (i) $\mathbb{E}[\epsilon_i] = 0$ and (ii) $\mathbb{E}[\epsilon_i \mathbf{X}_i] = \mathbf{0}$
 - $\mathbb{E}[\epsilon_i] = \mathbb{E}[\mathbb{E}[\epsilon_i | \mathbf{X}_i]] = \mathbb{E}[0] = 0$
 - $\mathbb{E}[\epsilon_i \mathbf{X}_i] = \mathbb{E}[\mathbb{E}[\epsilon_i \mathbf{X}_i | \mathbf{X}_i]] = \mathbb{E}[\mathbf{X}_i \mathbb{E}[\epsilon_i | \mathbf{X}_i]] = \mathbb{E}[\mathbf{X}_i \cdot 0] = \mathbf{0}$
- Thus, the error ϵ_i is uncorrelated with regressor $\mathbf{X}_i$
 - $\text{Cov}(\mathbf{X}_i, \epsilon_i) = \mathbb{E}[\epsilon_i \mathbf{X}_i] - \mathbb{E}[\epsilon_i] \mathbb{E}[\mathbf{X}_i] = 0$

Review: Projection Coefficient

- Consider minimizing the mean squared prediction error

$$\begin{aligned} S(\beta) &= \mathbb{E}[(Y_i - \mathbf{X}_i^\top \beta)^2] \\ &= \mathbb{E}[Y_i^2] - 2\beta^\top \mathbb{E}[\mathbf{X}_i Y_i] + \beta^\top \mathbb{E}[\mathbf{X}_i \mathbf{X}_i^\top] \beta \end{aligned}$$

- Taking the derivative, we get

$$\frac{\partial S(\beta)}{\partial \beta} = -2\mathbb{E}[\mathbf{X}_i Y_i] + 2\mathbb{E}[\mathbf{X}_i \mathbf{X}_i^\top] \beta$$

- This gives the projection coefficient β

$$\beta = (\mathbb{E}[\mathbf{X}_i \mathbf{X}_i^\top])^{-1} \mathbb{E}[\mathbf{X}_i Y_i]$$

- Check this by inserting $Y_i = \mathbf{X}_i^\top \beta + \epsilon$.

Why linear regression? Why projection?

- Linear regression is a **best linear approximation** of conditional expectation function $m(\mathbf{X}_i) = \mathbb{E}[Y_i | \mathbf{X}_i]$
- To see this, consider minimizing the mean squared loss between $m(\mathbf{X}_i)$ and $\mathbf{X}_i^\top \beta$

$$d(\beta) = \mathbb{E}[(m(\mathbf{X}_i) - \mathbf{X}_i^\top \beta)^2]$$

- Notice that this minimization problem is just replacing Y_i with $m(\mathbf{X}_i)$ from the previous one. Thus, the best linear approximation $\tilde{\beta}$ is given by

$$\begin{aligned}\tilde{\beta} &= (\mathbb{E}[\mathbf{X}_i \mathbf{X}_i^\top])^{-1} \mathbb{E}[\mathbf{X}_i m(\mathbf{X}_i)] \\ &= (\mathbb{E}[\mathbf{X}_i \mathbf{X}_i^\top])^{-1} \mathbb{E}[\mathbf{X}_i \mathbb{E}[Y_i | \mathbf{X}_i]] \\ &= (\mathbb{E}[\mathbf{X}_i \mathbf{X}_i^\top])^{-1} \mathbb{E}[\mathbb{E}[\mathbf{X}_i Y_i | \mathbf{X}_i]] \\ &= (\mathbb{E}[\mathbf{X}_i \mathbf{X}_i^\top])^{-1} \mathbb{E}[\mathbf{X}_i Y_i] = \beta\end{aligned}$$

Review: Ordinary Least Squares (OLS)

- Let's use matrix notation.

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \mathbf{X} = \begin{bmatrix} 1 & X_{11} & \cdots & X_{K1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & \cdots & X_{Kn} \end{bmatrix}, \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_K \end{bmatrix}$$

- OLS estimator is given by

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} = \left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i Y_i \right)$$

- This estimator is unbiased (Hint: show $\mathbb{E}[\hat{\boldsymbol{\beta}} \mid \mathbf{X}] = \boldsymbol{\beta}$)
 - Once you show $\mathbb{E}[\hat{\boldsymbol{\beta}} \mid \mathbf{X}] = \boldsymbol{\beta}$, then use the law of iterated expectation!
- Also, residual is orthogonal: $\mathbf{X}^\top \hat{\boldsymbol{\epsilon}} = \mathbf{X}^\top (\mathbf{Y} - \mathbf{X} \hat{\boldsymbol{\beta}}) = 0$
 - Hint: Insert the definition of $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$

Frisch-Waugh-Lowell (FWL) theorem

- Consider multiple regressions with K regressors

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_K X_{Ki} + \epsilon_i, \quad \mathbb{E}[\epsilon_i \mid X_{1i}, \dots, X_{Ki}] = 0$$

- Estimator for β_k needs matrix $\rightarrow$ often hard to make a proof
- FWL theorem** gives another formula for k -th coefficient

$$\beta_k = \frac{\text{Cov}(Y_i, \tilde{X}_{ki})}{\mathbb{V}[\tilde{X}_{ki}]} \quad \text{for } k \in \{1, \dots, K\}$$

where $\tilde{X}_{ik}$ is the residual obtained by

$$X_{ki} = \gamma_0 + \sum_{j \neq k} \gamma_j X_{ji} + \tilde{X}_{ki}$$

Frisch-Waugh-Lowell (FWL) theorem: Proof

$$\begin{aligned}\frac{\text{Cov}(Y_i, \tilde{X}_{ki})}{\mathbb{V}[\tilde{X}_{ki}]} &= \frac{\text{Cov}(\beta_0 + \beta_1 X_{1i} + \cdots + \beta_K X_{Ki} + \epsilon_i, \tilde{X}_{ki})}{\mathbb{V}[\tilde{X}_{ki}]} \quad (\because \text{def of } Y_i) \\ &= \frac{\text{Cov}(\beta_k X_{ki}, \tilde{X}_{ki})}{\mathbb{V}[\tilde{X}_{ki}]} \\ &= \frac{\text{Cov}(\beta_k \{\gamma_0 + \sum_{j \neq k} \gamma_j X_{ji} + \tilde{X}_{ki}\}, \tilde{X}_{ki})}{\mathbb{V}[\tilde{X}_{ki}]} \quad (\because \text{def of } X_{ki}) \\ &= \frac{\text{Cov}(\beta_k \tilde{X}_{ki}, \tilde{X}_{ki})}{\mathbb{V}[\tilde{X}_{ki}]} \quad (\because \text{Cov}(\tilde{X}_{ki}, X_{ji}) = 0 \text{ for } j \neq k) \\ &= \beta_k\end{aligned}$$

where the second line is because

- $\text{Cov}(\tilde{X}_{ki}, X_{ji}) = 0$ for any $j \neq k$
- $\text{Cov}(X_{li}, \epsilon) = 0$ for all $l \in \{1, \dots, K\} \rightarrow \text{Cov}(\tilde{X}_{ki}, \epsilon) = 0$

Frisch-Waugh-Lowell (FWL) theorem: Remark

- Note that you can also residualize the outcome; i.e.,

$$\beta_k = \frac{\text{Cov}(\tilde{Y}_i, \tilde{X}_{ki})}{\mathbb{V}[\tilde{X}_{ki}]} \quad \text{for } k \in \{1, \dots, K\}$$

where $\tilde{Y}_i$ is the residual obtained by

$$Y_i = \alpha_0 + \sum_{j \neq k} \alpha_j X_{ji} + \tilde{Y}_i$$

- FWL Theorem is a tool for you to make the proof of multiple regressions without using matrix algebra.
 - This can be a helpful tool for exam / problem set

Added: FWL Theorem with Residualized Outcome (1)

- **Goal:** Show that $\text{Cov}(\tilde{Y}_i, \tilde{X}_{ki}) = \text{Cov}(Y_i, \tilde{X}_{ki})$
 - Then the Remark follows directly from the FWL theorem
- **Key fact:** $\tilde{X}_{ki}$ is the residual from projecting X_{ki} on 1 and X_{ji} ($j \neq k$).
 - Thus,

$$\mathbb{E}[\tilde{X}_{ki}] = 0, \quad \mathbb{E}[X_{ji}\tilde{X}_{ki}] = 0 \quad \Rightarrow \quad \text{Cov}(X_{ji}, \tilde{X}_{ki}) = 0 \quad \text{for all } j \neq k$$

- Substitute the definition of $\tilde{Y}_i$ from the Remark and use the linearity of covariance:

$$\begin{aligned}\text{Cov}(\tilde{Y}_i, \tilde{X}_{ki}) &= \text{Cov}\left(Y_i - \alpha_0 - \sum_{j \neq k} \alpha_j X_{ji}, \tilde{X}_{ki}\right) \\ &= \text{Cov}(Y_i, \tilde{X}_{ki}) - \text{Cov}(\alpha_0, \tilde{X}_{ki}) - \sum_{j \neq k} \alpha_j \text{Cov}(X_{ji}, \tilde{X}_{ki})\end{aligned}$$

Added: FWL Theorem with Residualized Outcome (2)

- Both subtracted terms are zero (α_0 is a constant, and the key fact for each X_{ji}):

$$\begin{aligned}\text{Cov}(\tilde{Y}_i, \tilde{X}_{ki}) &= \text{Cov}(Y_i, \tilde{X}_{ki}) - \underbrace{\text{Cov}(\alpha_0, \tilde{X}_{ki})}_{=0} - \sum_{j \neq k} \alpha_j \underbrace{\text{Cov}(X_{ji}, \tilde{X}_{ki})}_{=0} \\ &= \text{Cov}(Y_i, \tilde{X}_{ki})\end{aligned}$$

- Divide both sides by $\mathbb{V}[\tilde{X}_{ki}]$ and apply the FWL theorem:

$$\frac{\text{Cov}(\tilde{Y}_i, \tilde{X}_{ki})}{\mathbb{V}[\tilde{X}_{ki}]} = \frac{\text{Cov}(Y_i, \tilde{X}_{ki})}{\mathbb{V}[\tilde{X}_{ki}]} = \beta_k$$

Asymptotic Variance (1)

- **Goal:** To understand where sandwich comes from.
- Recall that the estimator for $\hat{\beta}$ is given by

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} = \left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i Y_i \right)$$

- Because $\mathbf{Y} = \mathbf{X}\beta + \epsilon$,

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{X}\beta + \epsilon) = \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \epsilon$$

Thus,

$$\sqrt{n}(\hat{\beta} - \beta) = \sqrt{n}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \epsilon = \left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \right)^{-1} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{x}_i \epsilon_i \right)$$

Tools for Asymptotic Variance

- **Law of Large Numbers (LLN):** If $X_1, \dots, X_n$ are i.i.d.,

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \mathbb{E}[X_i]$$

- **Central Limit Theorem (CLT):** If $X_1, \dots, X_n$ are i.i.d.,

$$\sqrt{n}(\bar{X} - \mathbb{E}[X_i]) \xrightarrow{d} \mathcal{N}(0, \mathbb{V}[X_i])$$

- **Slutsky's Lemma:** If $X_n \xrightarrow{d} X$ for some random variable X and $Y_n \xrightarrow{P} c$ for some constant c ,

$$X_n Y_n \xrightarrow{d} cX$$

Asymptotic Variance (2)

- Notice that by law of large numbers, the first term converges in probability to

$$\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \xrightarrow{p} \mathbb{E}[\mathbf{x}^\top \mathbf{x}]$$

- On the other hand, because each $\mathbf{x}_{i\epsilon_i}$ are i.i.d., by central limit theorem, the second term converges in distribution to

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{x}_{i\epsilon_i} = \underbrace{\sqrt{n} \frac{1}{n} \sum_{i=1}^n \mathbf{x}_{i\epsilon_i}}_{\text{Form of Avg!}} \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[(\mathbf{x}^\top \boldsymbol{\epsilon} (\mathbf{x}^\top \boldsymbol{\epsilon})^\top)])$$

- Note that mean of normal distribution here is 0 because $\mathbb{E}[\mathbf{x}_{i\epsilon_i}] = 0$

Asymptotic Variance (3)

- Therefore,

$$\begin{aligned}\sqrt{n}(\hat{\beta} - \beta) &= \sqrt{n}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \epsilon \\ &\rightarrow \mathcal{N}\left(0, \underbrace{\mathbb{E}[\mathbf{X}^\top \mathbf{X}]^{-1} \mathbb{E}[\mathbf{X}^\top \epsilon \epsilon^\top \mathbf{X}] \mathbb{E}[\mathbf{X}^\top \mathbf{X}]^{-1}}_{\text{Asymptotic Variance}}\right)\end{aligned}$$

- If you assume homoskedasticity, the asymptotic variance is simplified

- **Homoskedasticity:** $\mathbb{V}[\epsilon_i | \mathbf{X}_i] = \sigma^2$, which gives

$$\mathbb{V}[\epsilon_i | \mathbf{X}_i] = \mathbb{E}[\epsilon_i^2 | \mathbf{X}_i] - \underbrace{\mathbb{E}[\epsilon_i | \mathbf{X}_i]^2}_{=0} = \mathbb{E}[\epsilon_i^2 | \mathbf{X}_i]$$

$$\begin{aligned}\Rightarrow \mathbb{E}[\mathbf{X} \epsilon \epsilon^\top \mathbf{X}^\top] &= \mathbb{E}[\mathbb{E}[\mathbf{X} \epsilon \epsilon^\top \mathbf{X}^\top | \mathbf{X}]] \quad (\because \text{Law of Iterated Expectation}) \\ &= \mathbb{E}[\mathbf{X} \underbrace{\mathbb{E}[\epsilon \epsilon^\top | \mathbf{X}]}_{=\sigma^2 I_n} \mathbf{X}^\top] = \sigma^2 \mathbb{E}[\mathbf{X}^\top \mathbf{X}]\end{aligned}$$

and thus the asymptotic variance is simplified to

$$\mathbb{E}[\mathbf{X}^\top \mathbf{X}]^{-1} \mathbb{E}[\mathbf{X}^\top \epsilon \epsilon^\top \mathbf{X}] \mathbb{E}[\mathbf{X}^\top \mathbf{X}]^{-1} = \sigma^2 \mathbb{E}[\mathbf{X}^\top \mathbf{X}]^{-1}$$

Eicker-Huber-White (EHW) robust variance estimator

- **Eicker-Huber-White (EHW) robust variance estimator** (or HC0)

$$\underbrace{(\mathbf{X}^\top \mathbf{X})^{-1}}_{\text{Bread}} \underbrace{\mathbf{X}^\top \text{diag}(\hat{\epsilon}_i^2) \mathbf{X}}_{\text{Meat}} \underbrace{(\mathbf{X}^\top \mathbf{X})^{-1}}_{\text{Bread}}$$

- Asymptotically consistent
 - i.e., as N increases, it approaches to the true value
- But this has some bias in finite sample
 - We will see it with simulation (two slides later)

Extra slides: HC2 variance estimator

- **HC2 heteroskedasticity-robust variance estimator**
 - Make finite sample bias smaller

$$\underbrace{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top}_{\text{Bread}} \underbrace{\text{diag}\left(\frac{\hat{\epsilon}_i^2}{1 - p_{ii}}\right) \mathbf{X}}_{\text{Meat}} \underbrace{(\mathbf{X}^\top \mathbf{X})^{-1}}_{\text{Bread}}$$

where $p_{ii} = \mathbf{X}_i(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}_i$

- When $\mathbf{X}_i = [1, T_i]^\top$ (i.e., no control),

$$\begin{aligned} p_{ii} &= \begin{bmatrix} 1 \\ T_i \end{bmatrix} \left(\begin{bmatrix} \sum_{i=1}^n 1 & \sum_{i=1}^n T_i \\ \sum_{i=1}^n T_i & \sum_{i=1}^n T_i^2 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 \\ T_i \end{bmatrix} \\ &= \frac{1}{n_1 n_0} \left(n_1(1 - T_i) + n_0 T_i \right) = \begin{cases} 1/n_1 & (\text{if } T_i = 1) \\ 1/n_0 & (\text{if } T_i = 0) \end{cases} \end{aligned}$$

- **Takeaway:** HC2 becomes Neyman Variance with no control

Extra slides: Many Robust Standard Errors

- Which robust SEs should we use in practice?
 - Do not use homoskedastic one unless you check it
 - When sample size is large, all robust SEs are consistent
 - For small sample, use degrees of freedom adjustment from Bell and McCaffrey (2002)
- Simulation results (from ECON2110 Lecture 12 Slide)

	$N = 1000$	$N = 100$	$N = 50$	$N = 20$
Homoskedasticity	91.72%	90.77%	90.07%	88.75%
HC1	95.42%	93.68%	92.17%	88.62%
HC2	95.45%	93.97%	92.81%	90.38%
HC2 w/ dof adjust	95.51%	94.8%	94.58%	96.38%

- Homoskedasticity → Does not have a proper coverage (95%)
- HC1 / HC2 → proper coverage for large sample, but some finite-sample bias

Regression and Causal Inference: No Control

- Consider the simple linear regression

$$Y_i = \alpha + \beta T_i + \epsilon_i \quad \text{equivalently} \quad \mathbb{E}[Y_i \mid T_i = t] = \alpha + \beta t$$

- Under complete randomization and consistency,

$$\mathbb{E}[Y_i \mid T_i = t] = \mathbb{E}[Y_i(t) \mid T_i = t] = \mathbb{E}[Y_i(t)]$$

- The regression above is written as

$$\mathbb{E}[Y_i(t)] = \alpha + \beta t$$

- Thus

$$\alpha = \mathbb{E}[Y_i(0)], \quad \beta = \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)] \quad (= \text{Average Treatment Effect})$$

- Takeaway:** Regression estimates ATE (same as diff-in-means)
 - Linearity does not matter
 - HC2 variance is numerical identical to Neyman's variance

No Interaction model: Consistency (1)

- After randomization, there might be imbalance in covariates
 - In this case, covariate adjustment helps improving efficiency
- Consider the multiple regression with demeaned pre-treatment covariates

$$Y_i = \alpha + \beta T_i + \gamma^\top \tilde{\mathbf{X}}_i + \epsilon_i$$

- OLS estimator minimizes the mean squared error

$$\begin{aligned} & \sum_{i=1}^n \{Y_i - \alpha - \beta T_i - \gamma^\top \tilde{\mathbf{X}}_i\}^2 \\ &= \sum_{i=1}^n \{Y_i - \alpha - \beta T_i\}^2 + \sum_{i=1}^n \{\gamma^\top \tilde{\mathbf{X}}_i\}^2 - 2 \sum_{i=1}^n \{(Y_i - \alpha - \beta T_i) \gamma^\top \tilde{\mathbf{X}}_i\} \end{aligned}$$

No Interaction model: Consistency (2)

- Notice that the last term converges to

$$\begin{aligned}\mathbb{E}[(Y_i - \alpha - \beta T_i)\gamma^\top \tilde{\mathbf{X}}_i] &= \mathbb{E}[Y_i \gamma^\top \tilde{\mathbf{X}}_i] - \alpha \underbrace{\gamma^\top \mathbb{E}[\tilde{\mathbf{X}}_i]}_{=0} - \beta \underbrace{\mathbb{E}[T_i \gamma^\top \tilde{\mathbf{X}}_i]}_{=\mathbb{E}[T_i] \gamma^\top \mathbb{E}[\tilde{\mathbf{X}}_i]=0} \\ &= \mathbb{E}[Y_i \gamma^\top \tilde{\mathbf{X}}_i]\end{aligned}$$

where $\mathbb{E}[T_i \mathbf{X}_i] = \mathbb{E}[T_i] \mathbb{E}[\tilde{\mathbf{X}}_i]$ by randomization of treatment and $\mathbb{E}[\tilde{\mathbf{X}}_i] = 0$ thanks to demeaning.

- So, the parameter of interest β only depends on the first term
 - Specification of covariates $\gamma^\top \tilde{\mathbf{X}}_i$ does not matter
- **Takeaway:** OLS estimator is consistent for PATE even if model is incorrect

No Interaction model: Efficiency

- Theorem 7.1 of Imbens and Rubin (2015):

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}\left(0, \frac{\mathbb{E}[(T_i - p)^2(Y_i - \alpha - \beta T_i - \gamma^\top \tilde{\mathbf{X}}_i)^2]}{p^2(1-p)^2}\right)$$

where $p = \frac{n_1}{n} = \mathbb{P}(T_i = 1)$

- If covariates predict outcome well, $Y_i - \alpha - \beta T_i - \gamma^\top \tilde{\mathbf{X}}_i$ becomes smaller
- **Takeaway:** Controlling $\tilde{\mathbf{X}}$ improves efficiency if model is correct
 - Efficiency gains can be lost under misspecifications (Freedman, 2008)
 - Though in most cases efficiency improves...

Fully-interacted model (1)

- Fully-interacted model gives you the imputation estimator for ATE
- Consider the model with interaction

$$Y_i = \alpha + \beta T_i + \gamma^\top \tilde{\mathbf{X}}_i + \delta^\top T_i \tilde{\mathbf{X}}_i + \epsilon_i$$

- Notice that

$$\mathbb{E}[Y_i(1) \mid \tilde{\mathbf{X}}_i] = \mathbb{E}[Y_i \mid T_i = 1, \tilde{\mathbf{X}}_i] = \alpha + \beta + (\gamma + \delta)^\top \tilde{\mathbf{X}}_i$$

$$\mathbb{E}[Y_i(0) \mid \tilde{\mathbf{X}}_i] = \mathbb{E}[Y_i \mid T_i = 0, \tilde{\mathbf{X}}_i] = \alpha + \gamma^\top \tilde{\mathbf{X}}_i$$

where $\mathbb{E}[Y_i(t) \mid \tilde{\mathbf{X}}_i] = \mathbb{E}[Y_i \mid T_i = t, \tilde{\mathbf{X}}_i]$ by complete randomization

- More specifically, we use $Y_i(t) \perp\!\!\!\perp T_i \mid \mathbf{X}_i$, which holds because

$$\begin{aligned}\mathbb{P}(Y_i(t) \mid T_i = t, \mathbf{X}_i) &= \frac{\mathbb{P}(Y_i(t), T_i = t, \mathbf{X}_i)}{\mathbb{P}(T_i = t, \mathbf{X}_i)} \\ &= \frac{\mathbb{P}(Y_i(t), \mathbf{X}_i) \mathbb{P}(T_i = t)}{\mathbb{P}(T_i = t) \mathbb{P}(\mathbf{X}_i)} = \mathbb{P}(Y_i(t) \mid \mathbf{X}_i)\end{aligned}$$

Fully-interacted model (2): Imputation Estimator

- Consider treated unit (i.e., $T_i = 1$). Then, $Y_i = Y_i(1)$, and $Y_i(0)$ is missing.
 - But we can impute $\widehat{Y_i(0)}$ using the fully interacted model
- In this case, the estimated treatment effect for that unit (τ_i) is written as

$$\hat{\tau}_i = Y_i(1) - \widehat{Y_i(0)} = Y_i - \hat{\alpha} - \hat{\gamma}^\top \tilde{\mathbf{X}}_i$$

- On the other hand, for the control unit,

$$\hat{\tau}_i = \widehat{Y_i(1)} - Y_i(0) = \hat{\alpha} + \hat{\beta} + \hat{\gamma}^\top \tilde{\mathbf{X}}_i + \hat{\delta}^\top \tilde{\mathbf{X}}_i - Y_i$$

- $\tilde{\mathbf{X}}_i$ is demeaned, so the average of imputation is consistent for each potential outcome regardless of model specification.

Fully-interacted model (3): Imputation Estimator

- Thus, the ATE estimator is written as

$$\begin{aligned} \frac{1}{N} \sum_{i=1}^n \hat{\tau}_i &= \overbrace{\frac{1}{N} \sum_{i=1}^n \left\{ T_i \left(Y_i(1) - \widehat{Y_i(0)} \right) + (1 - T_i) \left(\widehat{Y_i(1)} - Y_i(0) \right) \right\}}^{\text{Imputation Estimator}} \\ &= \frac{1}{N} \sum_{i=1}^n \left\{ T_i (Y_i - \hat{\alpha} - \hat{\gamma}^\top \tilde{\mathbf{X}}_i) + (1 - T_i) (\hat{\alpha} + \hat{\beta} + \hat{\gamma}^\top \tilde{\mathbf{X}}_i + \hat{\delta}^\top \tilde{\mathbf{X}}_i - Y_i) \right\} \end{aligned}$$

- You can show $\frac{1}{N} \sum_{i=1}^n \hat{\tau}_i = \hat{\beta}$ (below I show it)
 - Only takeaway is “regression gives you imputation estimator”
- Recall that residuals are orthogonal with regressor, which suggests

$$\underbrace{\begin{bmatrix} 1 & T_1 & \tilde{\mathbf{X}}_1 & T_1 \tilde{\mathbf{X}}_1 \\ \vdots & & & \vdots \\ 1 & T_n & \tilde{\mathbf{X}}_n & T_n \tilde{\mathbf{X}}_n \end{bmatrix}^\top}_{=\text{Transpose of Regressor}} \underbrace{\begin{bmatrix} Y_1 - \hat{\alpha} - \hat{\beta} T_1 - \hat{\gamma}^\top \tilde{\mathbf{X}}_1 - \hat{\delta}^\top T_1 \tilde{\mathbf{X}}_1 \\ \vdots \\ Y_n - \hat{\alpha} - \hat{\beta} T_n - \hat{\gamma}^\top \tilde{\mathbf{X}}_n - \hat{\delta}^\top T_n \tilde{\mathbf{X}}_n \end{bmatrix}}_{=\text{Residuals}} = 0$$

Fully-interacted model (4): Imputation Estimator

- The previous matrix calculation gives you

$$\begin{cases} \sum_{i=1}^n (Y_i - \hat{\alpha} - \hat{\beta} T_i - \hat{\gamma}^\top \tilde{\mathbf{X}}_i - \hat{\delta}^\top T_i \tilde{\mathbf{X}}_i) = 0 \\ \sum_{i=1}^n T_i (Y_i - \hat{\alpha} - \hat{\beta} T_i - \hat{\gamma}^\top \tilde{\mathbf{X}}_i - \hat{\delta}^\top T_i \tilde{\mathbf{X}}_i) = 0 \end{cases}$$

- The previous two formulas give you

$$\sum_{i=1}^n (1 - T_i) (Y_i - \hat{\alpha} - \hat{\beta} T_i - \hat{\gamma}^\top \tilde{\mathbf{X}}_i - \hat{\delta}^\top T_i \tilde{\mathbf{X}}_i) = 0$$

- Therefore, by noticing $T_i(1 - T_i) = 0$,

$$\begin{aligned} \sum_{i=1}^n T_i (Y_i - \hat{\alpha} - \hat{\beta} - \hat{\gamma}^\top \tilde{\mathbf{X}}_i - \hat{\delta}^\top \tilde{\mathbf{X}}_i) &= 0 \\ \sum_{i=1}^n (1 - T_i) (Y_i - \hat{\alpha} - \hat{\gamma}^\top \tilde{\mathbf{X}}_i) &= 0 \end{aligned}$$

Fully-interacted model (5): Imputation Estimator

- Using these equalities, the expression above simplifies to

$$\frac{1}{N} \sum_{i=1}^N \hat{\tau}_i = \frac{1}{N} \sum_{i=1}^N (\hat{\beta} + \hat{\delta}^\top \tilde{\mathbf{X}}_i) = \hat{\beta}$$

where the last equality is because $\tilde{\mathbf{X}}_i$ is demeaned.

- We did not use the fact that δ is correctly specified, so this is true even if the model is misspecified.
- Takeaway:** Fully-interacted model gives you imputation estimator
 - In Extra slides at the end, I show that they are at least as efficient as difference-in-means estimator using this expression
 - You can similarly show that it can be interpreted as projection estimator

$$\hat{\beta} = \frac{1}{N} \sum_{i=1}^N \left\{ \widehat{Y_i(1)} - \widehat{Y_i(0)} \right\}$$

- These are about the interpretation of $\hat{\beta}$ (coefficient in regression)

Fully-interacted model (6): Misspecification

- **Takeaway 1:** $\hat{\beta}$ from fully-interacted model is consistent for PATE even if model is incorrect
 - You can prove it using the same logic as in no-interaction case
- **Takeaway 2:** $\hat{\beta}$ from fully-interacted model is at least as efficient as the difference-in-means estimator even if model is incorrect
 - In `estimatr` package, there is `lm_lin` function for the implementation
- But these are under complete randomization!

Regression Adjustment under Stratified Design

- **Stratified Design:** Randomize treatment *within* strata
 - We have so far assumed complete randomization
 - Complete randomization $\rightarrow$ covariates and treatment are independent
 - Under stratified design, strata and treatment can be correlated
- Linear model with strata fixed effects

$$Y_i = \alpha_{\text{strata}_i} + \beta T_i + \epsilon_i$$

- The estimator $\hat{\beta}$ converges to the weighted average of strata specific ATE:

$$\hat{\beta} \xrightarrow{p} \frac{\sum_{j=1}^J \overbrace{w_j}^{\text{Strata } j\text{'s weight}} \cdot \overbrace{\mathbb{E}[Y_i(1) - Y_i(0) \mid \text{strata}_i = j]}^{\text{Strata } j\text{'s ATE}}}{\sum_{j=1}^J w_j k_j (1 - k_j)}$$

where $w_j = n_j/n$ and $k_j = n_{j1}/n_j$

- $\hat{\beta}$ converges to PATE either (i) when k_j is identical across strata or (ii) when strata-specific ATE is identical across strata

Stratified Design: FWL Derivation (1)

- **Goal:** Derive the weighted formula on the previous slide using the FWL theorem
 - FWL makes it transparent where the weight $k_j(1 - k_j)$ comes from
- Model with a full set of stratum fixed effects:

$$Y_i = \alpha_{\text{strata}_i} + \beta T_i + \epsilon_i = \sum_{j=1}^J \alpha_j D_{ij} + \beta T_i + \epsilon_i, \quad D_{ij} = \mathbb{1}\{\text{strata}_i = j\}$$

- Notation within stratum j (write $i \in j$ if $\text{strata}_i = j$):
 - n_j units: $n_{j1} = n_j k_j$ treated and $n_{j0} = n_j(1 - k_j)$ control
 - $\bar{Y}_j$: mean of Y_i in stratum j
 - $\bar{Y}_{1j}, \bar{Y}_{0j}$: mean of Y_i among treated / control units in stratum j

Stratified Design: FWL Derivation (2)

- **Step 1:** Residualize T_i and Y_i with respect to strata
- Regress T_i on the stratum dummies. Each unit is in exactly one stratum, so

$$\sum_{i=1}^n \left(T_i - \sum_{j=1}^J a_j D_{ij} \right)^2 = \sum_{j=1}^J \sum_{i \in j} (T_i - a_j)^2$$

Each stratum is a separate problem $\rightarrow \hat{a}_j$ is the stratum mean:

$$\hat{a}_j = \bar{T}_j = \frac{1}{n_j} \sum_{i \in j} T_i = \frac{n_{j1}}{n_j} = k_j$$

- Thus, for unit i in stratum j , the residuals are

$$\tilde{T}_i = T_i - k_j, \quad \tilde{Y}_i = Y_i - \bar{Y}_j$$

where $\tilde{Y}_i$ follows from the same argument applied to Y_i

- Stratum fixed effects $\rightarrow$ only **within-stratum** variation is left

Stratified Design: FWL Derivation (3)

- Sample version of FWL (residualizing both, as in the FWL Remark):

$$\hat{\beta} = \frac{\sum_{i=1}^n \tilde{T}_i \tilde{Y}_i}{\sum_{i=1}^n \tilde{T}_i^2}$$

- Sample analog of $\text{Cov}(\tilde{Y}_i, \tilde{T}_i)/\mathbb{V}[\tilde{T}_i]$
- Residuals have mean zero, and the $1/n$ in the numerator and denominator cancels
- Break both sums up by stratum:

$$\hat{\beta} = \frac{\sum_{j=1}^J \sum_{i \in j} (T_i - k_j)(Y_i - \bar{Y}_j)}{\sum_{j=1}^J \sum_{i \in j} (T_i - k_j)^2}$$

Stratified Design: FWL Derivation (4)

- **Step 2:** Denominator within stratum j
- T_i is binary, so $T_i - k_j$ takes only two values:

$$T_i - k_j = \begin{cases} 1 - k_j & \text{if } T_i = 1 \quad (n_j k_j \text{ units}) \\ -k_j & \text{if } T_i = 0 \quad (n_j(1 - k_j) \text{ units}) \end{cases}$$

- Therefore,

$$\begin{aligned} \sum_{i \in j} (T_i - k_j)^2 &= n_j k_j (1 - k_j)^2 + n_j (1 - k_j) k_j^2 \\ &= n_j k_j (1 - k_j) \{ (1 - k_j) + k_j \} \\ &= n_j k_j (1 - k_j) \end{aligned}$$

- This is where $k_j(1 - k_j)$ comes from: $\frac{1}{n_j} \sum_{i \in j} (T_i - k_j)^2 = k_j(1 - k_j)$ is the **within-stratum variance of T_i**
 - Same identity as Review Question 2, within stratum j :
 $n_{j1} n_{j0} / n_j = n_j k_j (1 - k_j)$

Stratified Design: FWL Derivation (5)

- **Step 3:** Numerator within stratum j , $\sum_{i \in j} (T_i - k_j)(Y_i - \bar{Y}_j)$
- The residualized treatment sums to zero within each stratum:

$$\sum_{i \in j} (T_i - k_j) = \sum_{i \in j} T_i - n_j k_j = n_{j1} - n_{j1} = 0$$

- So the $\bar{Y}_j$ term drops out ($\bar{Y}_j$ is constant within stratum j):

$$\begin{aligned} \sum_{i \in j} (T_i - k_j)(Y_i - \bar{Y}_j) &= \sum_{i \in j} (T_i - k_j)Y_i - \underbrace{\bar{Y}_j \sum_{i \in j} (T_i - k_j)}_{=0} \\ &= \sum_{i \in j} (T_i - k_j)Y_i \end{aligned}$$

- Consistent with the FWL Remark: residualizing Y_i does not change $\hat{\beta}$

Stratified Design: FWL Derivation (6)

- Separate treated and control units ($T_i - k_j = 1 - k_j$ or $-k_j$):

$$\sum_{i \in j} (T_i - k_j) Y_i = (1 - k_j) \sum_{i \in j: T_i=1} Y_i - k_j \sum_{i \in j: T_i=0} Y_i$$

- By the definitions of the group means,

$$\sum_{i \in j: T_i=1} Y_i = n_j k_j \bar{Y}_{1j}, \quad \sum_{i \in j: T_i=0} Y_i = n_j (1 - k_j) \bar{Y}_{0j}$$

- Substituting,

$$\begin{aligned} \sum_{i \in j} (T_i - k_j) Y_i &= n_j k_j (1 - k_j) \bar{Y}_{1j} - n_j k_j (1 - k_j) \bar{Y}_{0j} \\ &= n_j k_j (1 - k_j) \hat{\tau}_j \end{aligned}$$

where $\hat{\tau}_j = \bar{Y}_{1j} - \bar{Y}_{0j}$ is the difference-in-means in stratum j

- Same factor $n_j k_j (1 - k_j)$ as in the denominator!

Stratified Design: FWL Derivation (7)

- Plug Steps 2 and 3 into the FWL formula:

$$\hat{\beta} = \frac{\sum_{j=1}^J n_j k_j (1 - k_j) \hat{\tau}_j}{\sum_{j=1}^J n_j k_j (1 - k_j)}$$

- Divide the numerator and denominator by n , with $w_j = n_j/n$:

$$\hat{\beta} = \frac{\sum_{j=1}^J w_j k_j (1 - k_j) \hat{\tau}_j}{\sum_{j=1}^J w_j k_j (1 - k_j)}$$

- This is an exact finite-sample identity
 - Pure algebra: no randomization and no probability limits used yet
 - $\hat{\beta}$ is a weighted average of the stratum-specific difference-in-means estimators

Stratified Design: FWL Derivation (8)

- Treatment is completely randomized within each stratum, so by Case 1 (no control) applied to stratum j ,

$$\hat{\tau}_j = \bar{Y}_{1j} - \bar{Y}_{0j} \xrightarrow{p} \tau_j = \mathbb{E}[Y_i(1) - Y_i(0) \mid \text{strata}_i = j]$$

- The weights w_j and k_j are fixed by the design, so

$$\hat{\beta} \xrightarrow{p} \frac{\sum_{j=1}^J w_j k_j (1 - k_j) \tau_j}{\sum_{j=1}^J w_j k_j (1 - k_j)}$$

- This is exactly the formula on the stratified design slide

Stratified Design: FWL Derivation (9): Intuition

- Stratum fixed effects turn the regression into a within-stratum regression
- But strata do not contribute equally. Stratum j 's weight is proportional to

$$\underbrace{n_j}_{\text{stratum size}} \times \underbrace{k_j(1 - k_j)}_{\text{within-stratum variation in } T_i}$$

- $k_j(1 - k_j)$ is largest when $k_j = 0.5$
 - $k_j = 0.5$: $k_j(1 - k_j) = 0.25 \rightarrow$ most treatment variation, most weight
 - $k_j = 0.05$: $k_j(1 - k_j) = 0.0475 \rightarrow$ less than 1/5 of the weight of a 50/50 stratum of the same size
- If outcomes are homoskedastic, $\mathbb{V}[\hat{\tau}_j] = \sigma^2 \left(\frac{1}{n_{j1}} + \frac{1}{n_{j0}} \right) = \frac{\sigma^2}{n_j k_j (1 - k_j)}$
 - OLS weights each stratum by the inverse of its variance

Stratified Design: FWL Derivation (10): When Is It PATE?

- By contrast, PATE weights strata only by their size:

$$\tau_{\text{PATE}} = \sum_{j=1}^J w_j \tau_j$$

- If $k_j = k$ for every stratum, $k(1 - k)$ cancels (and $\sum_{j=1}^J w_j = 1$):

$$\hat{\beta} \xrightarrow{p} \frac{k(1 - k) \sum_{j=1}^J w_j \tau_j}{k(1 - k) \sum_{j=1}^J w_j} = \sum_{j=1}^J w_j \tau_j = \tau_{\text{PATE}}$$

Similarly, if $\tau_j = \tau$ for every stratum, any weighted average of the τ_j 's equals $\tau = \tau_{\text{PATE}}$

Cluster Randomized Trials

- We assume *no interference* (implied by consistency)
 - Interference: potential outcome is function of other people's treatment status
- **Cluster randomized experiment:** assign treatment at the cluster level
 - We allow spillover *within* each cluster
 - We assume no spillover *across* clusters
- Notation
 - $j \in \{1, \dots, J\}$: cluster indicator
 - $i \in \{1, \dots, m_j\}$: individual indicator
 - T_j : treatment indicator for cluster j
 - Y_{ij} : outcome for individual i at cluster j
- Because everyone in each cluster is in the same treatment status,

$$\underbrace{Y_{ij}(T_{1j}, \dots, T_{m_jj})}_{\text{Allowing interference within cluster } j} = Y_{ij}(T_j)$$

Allowing interference within cluster j

Inference under Cluster Randomization

- Inference under clustering: regarding each cluster as unit

$$\text{ATE Estimator: } \hat{\tau}_{\text{cluster}} = \underbrace{\frac{1}{J_1} \sum_{j=1}^J T_j \bar{Y}_j}_{\text{Avg. of Treated Cluster}} - \underbrace{\frac{1}{J_0} \sum_{j=1}^J (1 - T_j) \bar{Y}_j}_{\text{Avg. of Control Cluster}}$$

$$\text{Variance Estimator: } \mathbb{V}[\widehat{\hat{\tau}_{\text{cluster}}}] = \frac{\text{var}(\bar{Y}_j(1))}{J_1} + \frac{\text{var}(\bar{Y}_j(0))}{J_0}$$

- These estimators look similar to Neyman's analysis

Efficiency Loss by Clustering

- If we assume $m_j = m$ for all j ,

$$\underbrace{\frac{\text{var}(\bar{Y}_j(t))}{J_t}}_{\text{Var w/ Clustering for } T_i=t} = \underbrace{\frac{\text{var}(Y_{ij}(t))}{J_t m}}_{\text{Variance w/o Clustering for } T_i=t} \underbrace{\left(1 + (m-1)\rho_t\right)}_{\geq 1 \text{ if ICC is positive}}$$

where ρ_t is ICC (Intra-cluster correlation coefficient) for treatment status t

- ICC is typically positive $\rightarrow$ Clustering typically loses efficiency
- We will derive this formula in the next few slides

Efficiency Loss by Clustering: Derivation (1)

- Recall that $\bar{Y}_j(t)$ is the average of m individuals in cluster j

$$\bar{Y}_j(t) = \frac{1}{m} \sum_{i=1}^m Y_{ij}(t)$$

- By linearity of expectation, $\mathbb{E}[\bar{Y}_j(t)] = \frac{1}{m} \sum_{i=1}^m \mathbb{E}[Y_{ij}(t)]$
- Thus,

$$\begin{aligned} \text{var}(\bar{Y}_j(t)) &= \mathbb{E}[(\bar{Y}_j(t) - \mathbb{E}[\bar{Y}_j(t)])^2] \quad (\because \text{def of variance}) \\ &= \mathbb{E}\left[\left\{\frac{1}{m} \sum_{i=1}^m Y_{ij}(t) - \frac{1}{m} \sum_{i=1}^m \mathbb{E}[Y_{ij}(t)]\right\}^2\right] \\ &= \frac{1}{m^2} \mathbb{E}\left[\left\{\sum_{i=1}^m (Y_{ij}(t) - \mathbb{E}[Y_{ij}(t)])\right\}^2\right] \end{aligned}$$

Efficiency Loss by Clustering: Derivation (2)

- Square of the sum = sum over all pairs (i, i')

$$\begin{aligned} & \left\{ \sum_{i=1}^m (Y_{ij}(t) - \mathbb{E}[Y_{ij}(t)]) \right\}^2 \\ &= \sum_{i=1}^m \sum_{i'=1}^m (Y_{ij}(t) - \mathbb{E}[Y_{ij}(t)])(Y_{i'j}(t) - \mathbb{E}[Y_{i'j}(t)]) \\ &= \underbrace{\sum_{i=1}^m (Y_{ij}(t) - \mathbb{E}[Y_{ij}(t)])^2}_{i'=i \text{ (} m \text{ terms)}} \\ & \quad + \underbrace{\sum_{i=1}^m \sum_{i' \neq i} (Y_{ij}(t) - \mathbb{E}[Y_{ij}(t)])(Y_{i'j}(t) - \mathbb{E}[Y_{i'j}(t)])}_{i' \neq i \text{ (} m(m-1) \text{ terms)}} \end{aligned}$$

Efficiency Loss by Clustering: Derivation (3)

- Taking expectation of each term (by def of variance and covariance)

$$\begin{aligned}\mathbb{E}[(Y_{ij}(t) - \mathbb{E}[Y_{ij}(t)])^2] &= \text{var}(Y_{ij}(t)) \\ \mathbb{E}[(Y_{ij}(t) - \mathbb{E}[Y_{ij}(t)])(Y_{i'j}(t) - \mathbb{E}[Y_{i'j}(t)])] &= \text{Cov}(Y_{ij}(t), Y_{i'j}(t))\end{aligned}$$

- Plugging these into Derivation (1)–(2) gives

$$\frac{\text{var}(\bar{Y}_j(t))}{J_t} = \frac{1}{J_t m^2} \left\{ \sum_{i=1}^m \text{var}(Y_{ij}(t)) + \sum_{i=1}^m \sum_{i' \neq i} \text{Cov}(Y_{ij}(t), Y_{i'j}(t)) \right\}$$

- If all covariances are 0, only the **red term** remains (no clustering penalty)

Efficiency Loss by Clustering: Derivation (4)

- For simplification, we assume that individuals in the same cluster are identically distributed
 - Same mean: $\mathbb{E}[Y_{ij}(t)] = \mu_t$ for all $i \in \{1, \dots, m\}$
 - Same variance: $\text{var}(Y_{ij}(t)) = \text{var}(Y_{i'j}(t))$ for all $i, i' \in \{1, \dots, m\}$
- Red term:

$$\sum_{i=1}^m \text{var}(Y_{ij}(t)) = m \text{var}(Y_{ij}(t)) \quad (\because \text{same variance})$$

- Blue term: for each pair $i' \neq i$,

$$\begin{aligned} \text{Cov}(Y_{ij}(t), Y_{i'j}(t)) &= \text{Corr}(Y_{ij}(t), Y_{i'j}(t)) \sqrt{\text{var}(Y_{ij}(t)) \text{var}(Y_{i'j}(t))} \\ &= \text{Corr}(Y_{ij}(t), Y_{i'j}(t)) \text{var}(Y_{ij}(t)) \end{aligned}$$

where the first line is the def of Corr and the second line uses same variance

Efficiency Loss by Clustering: Derivation (5)

- Summing over the $m(m-1)$ pairs,

$$\begin{aligned}\sum_{i=1}^m \sum_{i' \neq i} \text{Cov}(Y_{ij}(t), Y_{i'j}(t)) &= \text{var}(Y_{ij}(t)) \sum_{i=1}^m \sum_{i' \neq i} \text{Corr}(Y_{ij}(t), Y_{i'j}(t)) \\ &= m(m-1)\rho_t \text{var}(Y_{ij}(t))\end{aligned}$$

where $\rho_t = \frac{1}{m(m-1)} \sum_{i=1}^m \sum_{i' \neq i} \text{Corr}(Y_{ij}(t), Y_{i'j}(t))$ is the **average** correlation among all pairs in the cluster (**Intraclass Correlation Coefficient**)

- If you further assume that all pairs have the same correlation, then $\rho_t = \text{Corr}(Y_{ij}(t), Y_{i'j}(t))$

Efficiency Loss by Clustering: Derivation (6)

- Plugging both terms into the derived formula gives

$$\begin{aligned}\frac{\text{var}(\bar{Y}_j(t))}{J_t} &= \frac{1}{J_t m^2} \left\{ m \text{var}(Y_{ij}(t)) + m(m-1)\rho_t \text{var}(Y_{ij}(t)) \right\} \\ &= \frac{m \text{var}(Y_{ij}(t))}{J_t m^2} \{1 + (m-1)\rho_t\} \\ &= \frac{\text{var}(Y_{ij}(t))}{J_t m} \{1 + (m-1)\rho_t\}\end{aligned}$$

- Special cases

- $\rho_t = 0 \rightarrow$ same as Neyman variance with $J_t m$ people
- $\rho_t = 1 \rightarrow$ effectively only J_t observations
- $m = 101, \rho_t = 0.5 \rightarrow 1 + 100 \times 0.5 = 51$ times greater

- **Takeaway:** When ρ_t is large, adding people to each cluster helps little

Extra Slides: Cluster Robust Standard Errors (CR)

- As with the previous case,

$$\begin{aligned}\sqrt{n}(\hat{\beta} - \beta) &= \sqrt{n}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \epsilon \\ &= \left(\frac{1}{n} \sum_{j=1}^J \sum_{i=1}^{m_j} \mathbf{x}_{ij} \mathbf{x}_{ij}^\top \right)^{-1} \left(\frac{1}{\sqrt{n}} \sum_{j=1}^J \sum_{i=1}^{m_j} \mathbf{x}_{ij} \epsilon_{ij} \right)\end{aligned}$$

- Independence holds across clusters (not within clusters!)
- Cluster Robust Variance Estimator**

$$\underbrace{\left(\sum_{j=1}^J \mathbf{x}_j^\top \mathbf{x}_j \right)^{-1}}_{\text{Bread}} \underbrace{\left(\sum_{j=1}^J \mathbf{x}_j^\top \hat{\epsilon}_j \hat{\epsilon}_j^\top \mathbf{x}_j \right)}_{\text{Meat}} \underbrace{\left(\sum_{j=1}^J \mathbf{x}_j^\top \mathbf{x}_j \right)^{-1}}_{\text{Bread}}$$

- CR2** (Bias-adjustment): Same idea as HC2
 - Same degrees of freedom adjustment is available for small sample

Review Question: Setting

- We work in the **super-population framework**: the pairs $(Y_i(0), Y_i(1))$ are i.i.d. across units. Let

$$\mu_t = \mathbb{E}[Y_i(t)], \quad \sigma_t^2 = \mathbb{V}[Y_i(t)] < \infty, \quad t \in \{0, 1\}.$$

- Under complete randomization, n_1 and n_0 are fixed and $n = n_1 + n_0$. Assume $n_1, n_0 \geq 2$.
 - Treatment assignment is independent of all potential outcomes.
 - We allow $\sigma_1^2 \neq \sigma_0^2$ (**heteroskedasticity**).
- Write $\mathbf{T} = (T_1, \dots, T_n)^\top$ and $\mathcal{O}_n = \{Y_i(0), Y_i(1)\}_{i=1}^n$.
 - Given $\mathcal{O}_n$, only treatment assignment is random.
 - Given $\mathbf{T}$, potential outcomes are still random.
 - Unconditional expectations average over both sources of randomness.

Review Question: Setting (continued)

- Fit $Y_i = \alpha + \beta T_i + \epsilon_i$. Define

$$\bar{Y}_1 = \frac{1}{n_1} \sum_{i=1}^n T_i Y_i, \quad \bar{Y}_0 = \frac{1}{n_0} \sum_{i=1}^n (1 - T_i) Y_i,$$

$$\hat{Y}_i = \hat{\alpha} + \hat{\beta} T_i, \quad \hat{\epsilon}_i = Y_i - \hat{Y}_i, \quad \bar{T} = \frac{n_1}{n}.$$

- The **conventional OLS variance estimator** is

$$\hat{V}_{\text{OLS}} = \frac{\hat{\sigma}^2}{\sum_{i=1}^n (T_i - \bar{T})^2}, \quad \hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{\epsilon}_i^2.$$

The denominator $n - 2$ accounts for estimating α and β .

- This formula pools the residual variation in the two groups. Is it unbiased when $\sigma_1^2 \neq \sigma_0^2$? Let's check this using the three review questions.

Review Question 1

- Show that

$$\mathbb{E}[\hat{\sigma}^2] = \frac{(n_1 - 1)\sigma_1^2 + (n_0 - 1)\sigma_0^2}{n - 2}.$$

- Hint: center the observed and fitted outcomes at the same population mean:

$$C_{0i} = \{(Y_i - \mu_0) - (\hat{Y}_i - \mu_0)\}^2(1 - T_i),$$

$$C_{1i} = \{(Y_i - \mu_1) - (\hat{Y}_i - \mu_1)\}^2 T_i.$$

- First show

$$\mathbb{E} \left[\sum_{i=1}^n C_{ti} \right] = (n_t - 1)\sigma_t^2, \quad t \in \{0, 1\},$$

then add the two groups and divide by $n - 2$.

Review Question 1 - Answer

- First, find the OLS fitted values. The normal equations imply

$$\sum_{i=1}^n \hat{\epsilon}_i = 0, \quad \sum_{i=1}^n T_i \hat{\epsilon}_i = 0.$$

- The second equation says that treated residuals sum to zero:

$$\begin{aligned} 0 &= \sum_{i=1}^n T_i (Y_i - \hat{\alpha} - \hat{\beta} T_i) \\ &= n_1 \bar{Y}_1 - n_1 \hat{\alpha} - n_1 \hat{\beta} \quad (\because T_i^2 = T_i), \\ \hat{\alpha} + \hat{\beta} &= \bar{Y}_1. \end{aligned}$$

- Subtracting the second normal equation from the first gives

$$0 = \sum_{i=1}^n (1 - T_i) \hat{\epsilon}_i = n_0 \bar{Y}_0 - n_0 \hat{\alpha}.$$

Thus, $\hat{\alpha} = \bar{Y}_0$ and $\hat{\beta} = \bar{Y}_1 - \bar{Y}_0$.

Review Question 1 - Answer (2)

- Substitute the coefficients into the fitted outcome:

$$\hat{Y}_i = \bar{Y}_0 + (\bar{Y}_1 - \bar{Y}_0)T_i = \begin{cases} \bar{Y}_0 & \text{if } T_i = 0, \\ \bar{Y}_1 & \text{if } T_i = 1. \end{cases}$$

- Split the squared residuals using $(1 - T_i) + T_i = 1$:

$$\begin{aligned} \sum_{i=1}^n \hat{\epsilon}_i^2 &= \sum_{i=1}^n (1 - T_i)(Y_i - \hat{Y}_i)^2 + \sum_{i=1}^n T_i(Y_i - \hat{Y}_i)^2 \\ &= \sum_{i=1}^n C_{0i} + \sum_{i=1}^n C_{1i}. \end{aligned}$$

Subtracting μ_t from both Y_i and $\hat{Y}_i$ leaves their difference unchanged.

- Divide by $n - 2$ and take expectation:

$$\mathbb{E}[\hat{\sigma}^2] = \frac{1}{n-2} \left\{ \mathbb{E} \left[\sum_{i=1}^n C_{0i} \right] + \mathbb{E} \left[\sum_{i=1}^n C_{1i} \right] \right\}.$$

Review Question 1 - Answer (3)

- Consider the treated group first. Since $\hat{Y}_i = \bar{Y}_1$ when $T_i = 1$,

$$\sum_{i=1}^n C_{1i} = \sum_{i=1}^n T_i \{(Y_i - \mu_1) - (\bar{Y}_1 - \mu_1)\}^2.$$

- Expand the square using $(a - b)^2 = a^2 - 2ab + b^2$:

$$\begin{aligned} \sum_{i=1}^n C_{1i} &= \sum_{i=1}^n T_i (Y_i - \mu_1)^2 \\ &\quad - 2 \sum_{i=1}^n T_i (Y_i - \mu_1)(\bar{Y}_1 - \mu_1) \\ &\quad + \sum_{i=1}^n T_i (\bar{Y}_1 - \mu_1)^2. \end{aligned}$$

- $\bar{Y}_1 - \mu_1$ is the same for every i , so we can take it outside the sums. It is still a **random variable**!

Review Question 1 - Answer (4)

- Use the definitions of n_1 and $\bar{Y}_1$:

$$\sum_{i=1}^n T_i = n_1, \quad \sum_{i=1}^n T_i(Y_i - \mu_1) = n_1 \bar{Y}_1 - n_1 \mu_1 = n_1(\bar{Y}_1 - \mu_1).$$

- Substitute into the expanded square and combine the last two terms:

$$\begin{aligned} \sum_{i=1}^n C_{1i} &= \sum_{i=1}^n T_i(Y_i - \mu_1)^2 - 2n_1(\bar{Y}_1 - \mu_1)^2 + n_1(\bar{Y}_1 - \mu_1)^2 \\ &= \sum_{i=1}^n T_i(Y_i - \mu_1)^2 - n_1(\bar{Y}_1 - \mu_1)^2. \end{aligned}$$

- By linearity of expectation, we need to calculate two terms:

$$\mathbb{E} \left[\sum_{i=1}^n C_{1i} \right] = \underbrace{\sum_{i=1}^n \mathbb{E}[T_i(Y_i - \mu_1)^2]}_{\text{First term}} - n_1 \underbrace{\mathbb{E}[(\bar{Y}_1 - \mu_1)^2]}_{\text{Second term}}.$$

Review Question 1 - Answer (5)

- **First term:** $T_i Y_i = T_i Y_i(1)$ by consistency. Given $\mathcal{O}_n$, $Y_i(1)$ is fixed and $\mathbb{E}[T_i \mid \mathcal{O}_n] = n_1/n$.
- Apply the law of iterated expectation:

$$\begin{aligned}\sum_{i=1}^n \mathbb{E}[T_i(Y_i - \mu_1)^2] &= \sum_{i=1}^n \mathbb{E}[T_i(Y_i(1) - \mu_1)^2] \\ &= \sum_{i=1}^n \mathbb{E} \left[\mathbb{E}[T_i \mid \mathcal{O}_n](Y_i(1) - \mu_1)^2 \right] \\ &= \frac{n_1}{n} \sum_{i=1}^n \mathbb{E}[(Y_i(1) - \mu_1)^2] \\ &= \frac{n_1}{n} \cdot n\sigma_1^2 \\ &= n_1\sigma_1^2.\end{aligned}$$

- The fourth line uses $\mathbb{E}[(Y_i(1) - \mu_1)^2] = \sigma_1^2$.

Review Question 1 - Answer (6)

- **Second term:** first check that $\bar{Y}_1$ is unbiased for μ_1 . Random assignment is independent of potential outcomes:

$$\begin{aligned}\mathbb{E}[\bar{Y}_1] &= \frac{1}{n_1} \sum_{i=1}^n \mathbb{E}[T_i Y_i(1)] \\ &= \frac{1}{n_1} \sum_{i=1}^n \mathbb{E}[T_i] \mathbb{E}[Y_i(1)] \\ &= \frac{1}{n_1} \cdot n \cdot \frac{n_1}{n} \mu_1 = \mu_1.\end{aligned}$$

- Thus, by the definition of variance,

$$\mathbb{E}[(\bar{Y}_1 - \mu_1)^2] = \mathbb{V}[\bar{Y}_1] = \mathbb{E}[\bar{Y}_1^2] - \mu_1^2.$$

- We now calculate $\mathbb{E}[\bar{Y}_1^2]$ by expanding the square into terms with $i = j$ and $i \neq j$.

Review Question 1 - Answer (7)

- For the terms with $i = j$, use $T_i^2 = T_i$:

$$\mathbb{E}[T_i Y_i(1)^2] = \mathbb{E}[T_i] \mathbb{E}[Y_i(1)^2] = \frac{n_1}{n} (\sigma_1^2 + \mu_1^2).$$

- For $i \neq j$, treatment assignments are **dependent**:

$$\begin{aligned}\mathbb{E}[T_i T_j] &= \mathbb{P}(T_i = 1) \mathbb{P}(T_j = 1 \mid T_i = 1) \\ &= \frac{n_1}{n} \cdot \frac{n_1 - 1}{n - 1}.\end{aligned}$$

Treating one unit leaves $n_1 - 1$ treatment slots among $n - 1$ units.

- Potential outcomes are independent **across units**, so

$$\begin{aligned}\mathbb{E}[T_i T_j Y_i(1) Y_j(1)] &= \mathbb{E}[T_i T_j] \mathbb{E}[Y_i(1)] \mathbb{E}[Y_j(1)] \\ &= \frac{n_1(n_1 - 1)}{n(n - 1)} \mu_1^2.\end{aligned}$$

Review Question 1 - Answer (8)

- The square of a sum includes all ordered pairs (i, j) :

$$\begin{aligned}\mathbb{E}[\bar{Y}_1^2] &= \frac{1}{n_1^2} \mathbb{E} \left[\left(\sum_{i=1}^n T_i Y_i(1) \right)^2 \right] \\ &= \frac{1}{n_1^2} \left\{ \sum_{i=1}^n \mathbb{E}[T_i Y_i(1)^2] + \sum_{i \neq j} \mathbb{E}[T_i T_j Y_i(1) Y_j(1)] \right\}.\end{aligned}$$

- There are n diagonal terms and $n(n-1)$ off-diagonal terms. Substitute the two moments from the previous slide:

$$\begin{aligned}\mathbb{E}[\bar{Y}_1^2] &= \frac{1}{n_1^2} \left\{ n \frac{n_1}{n} (\sigma_1^2 + \mu_1^2) + n(n-1) \frac{n_1(n_1-1)}{n(n-1)} \mu_1^2 \right\} \\ &= \frac{\sigma_1^2 + \mu_1^2}{n_1} + \frac{n_1-1}{n_1} \mu_1^2.\end{aligned}$$

Review Question 1 - Answer (9)

- Subtract the squared mean and collect the terms in μ_1^2 :

$$\begin{aligned}\mathbb{V}[\bar{Y}_1] &= \mathbb{E}[\bar{Y}_1^2] - \mu_1^2 \\ &= \frac{\sigma_1^2 + \mu_1^2}{n_1} + \frac{n_1 - 1}{n_1} \mu_1^2 - \mu_1^2 \\ &= \frac{\sigma_1^2}{n_1} + \left(\frac{1 + n_1 - 1}{n_1} - 1 \right) \mu_1^2 \\ &= \frac{\sigma_1^2}{n_1}.\end{aligned}$$

- We will also use the conditional version in Question 3:

$$\mathbb{V}[\bar{Y}_t \mid \mathbf{T}] = \frac{1}{n_t^2} \sum_{i: T_i=t} \mathbb{V}[Y_i(t) \mid \mathbf{T}] = \frac{\sigma_t^2}{n_t}.$$

Given the assignment, each group consists of n_t independent observations from the super-population.

Review Question 1 - Answer (10)

- Return to the two terms from Answer 1 (4):

$$\begin{aligned}\mathbb{E}\left[\sum_{i=1}^n C_{1i}\right] &= \sum_{i=1}^n \mathbb{E}[T_i(Y_i - \mu_1)^2] - n_1 \mathbb{E}[(\bar{Y}_1 - \mu_1)^2] \\ &= n_1 \sigma_1^2 - n_1 \frac{\sigma_1^2}{n_1} \\ &= n_1 \sigma_1^2 - \sigma_1^2 \\ &= (n_1 - 1) \sigma_1^2.\end{aligned}$$

- Estimating the treated mean uses one degree of freedom. This is why we get $n_1 - 1$, rather than n_1 .

Review Question 1 - Answer (11)

- For the control group, replace T_i with $1 - T_i$ and (n_1, μ_1, σ_1^2) with (n_0, μ_0, σ_0^2) :

$$\begin{aligned}\mathbb{E}\left[\sum_{i=1}^n C_{0i}\right] &= \sum_{i=1}^n \mathbb{E}[(1 - T_i)(Y_i - \mu_0)^2] - n_0 \mathbb{E}[(\bar{Y}_0 - \mu_0)^2] \\ &= n_0 \sigma_0^2 - n_0 \frac{\sigma_0^2}{n_0} = (n_0 - 1) \sigma_0^2.\end{aligned}$$

- Add the two groups and divide by $n - 2$:

$$\begin{aligned}\mathbb{E}[\hat{\sigma}^2] &= \frac{1}{n-2} \left\{ \mathbb{E}\left[\sum_{i=1}^n C_{0i}\right] + \mathbb{E}\left[\sum_{i=1}^n C_{1i}\right] \right\} \\ &= \frac{(n_0 - 1) \sigma_0^2 + (n_1 - 1) \sigma_1^2}{n - 2}.\end{aligned}$$

- Takeaway:** $\hat{\sigma}^2$ estimates a weighted average of the two variances, with weights proportional to $n_t - 1$.

Review Question 2 (Optional)

- Show that the denominator of the OLS variance estimator is

$$\sum_{i=1}^n (T_i - \bar{T})^2 = \frac{n_1 n_0}{n}.$$

- Recall that

$$T_i^2 = T_i, \quad \sum_{i=1}^n T_i = n_1, \quad \bar{T} = \frac{n_1}{n}, \quad n = n_1 + n_0.$$

- This is an algebraic identity for a binary treatment variable. We do not need to take expectation.

Review Question 2 - Answer 2 (1)

- Expand the square and use $T_i^2 = T_i$:

$$\begin{aligned}\sum_{i=1}^n (T_i - \bar{T})^2 &= \sum_{i=1}^n \{T_i^2 - 2T_i\bar{T} + \bar{T}^2\} \\ &= \sum_{i=1}^n T_i - 2\bar{T} \sum_{i=1}^n T_i + \sum_{i=1}^n \bar{T}^2 \\ &= n_1 - 2\bar{T}n_1 + n\bar{T}^2.\end{aligned}$$

- $\bar{T}$ does not depend on i , so the last sum has n identical terms.
- Now substitute $\bar{T} = n_1/n$:

$$\sum_{i=1}^n (T_i - \bar{T})^2 = n_1 - 2n_1 \frac{n_1}{n} + n \left(\frac{n_1}{n} \right)^2.$$

Review Question 2 - Answer 2 (2)

- Simplify the fractions and factor out n_1 :

$$\begin{aligned}\sum_{i=1}^n (T_i - \bar{T})^2 &= n_1 - \frac{2n_1^2}{n} + \frac{n_1^2}{n} \\&= n_1 - \frac{n_1^2}{n} \\&= \frac{n_1 n - n_1^2}{n} \\&= \frac{n_1(n - n_1)}{n} \\&= \frac{n_1 n_0}{n} \quad (\because n - n_1 = n_0).\end{aligned}$$

- Under complete randomization, n_1, n_0 are fixed. Thus, the denominator of $\hat{V}_{OLS}$ is fixed, even though the individual T_i 's are random.

Review Question 3

- Finally, use the results from Questions 1 and 2 to derive the bias of the conventional OLS variance estimator:

$$\text{Bias}(\hat{V}_{\text{OLS}}) = \frac{(n_1 - n_0)(n - 1)(\sigma_1^2 - \sigma_0^2)}{n_1 n_0 (n - 2)}.$$

- Hint:** Bias is the expected estimate minus the true variance:

$$\text{Bias}(\hat{V}_{\text{OLS}}) = \mathbb{E}[\hat{V}_{\text{OLS}}] - \mathbb{V}[\hat{\beta}].$$

- Recall that we average over both random sampling from the super-population and treatment assignment.

Review Question 3 - Answer 3 (1)

- Conditional on $\mathbf{T}$, the two groups contain **different units**. I.i.d. sampling of potential-outcome pairs gives

$$\text{Cov}(\bar{Y}_1, \bar{Y}_0 \mid \mathbf{T}) = 0.$$

- Since $\hat{\beta} = \bar{Y}_1 - \bar{Y}_0$, the true variance is

$$\begin{aligned}\mathbb{V}[\hat{\beta} \mid \mathbf{T}] &= \mathbb{V}[\bar{Y}_1 - \bar{Y}_0 \mid \mathbf{T}] \\ &= \mathbb{V}[\bar{Y}_1 \mid \mathbf{T}] + \mathbb{V}[\bar{Y}_0 \mid \mathbf{T}] \\ &\quad - 2\text{Cov}(\bar{Y}_1, \bar{Y}_0 \mid \mathbf{T}) \\ &= \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}.\end{aligned}$$

- Also, $\mathbb{E}[\hat{\beta} \mid \mathbf{T}] = \mu_1 - \mu_0$ is constant. By the law of total variance, $\mathbb{V}[\hat{\beta}] = \mathbb{V}[\hat{\beta} \mid \mathbf{T}]$.

Review Question 3 - Answer 3 (2)

- First, substitute the denominator from Question 2:

$$\begin{aligned}\mathbb{E}[\widehat{V}_{\text{OLS}}] &= \mathbb{E}\left[\frac{\hat{\sigma}^2}{\sum_{i=1}^n (T_i - \bar{T})^2}\right] \\ &= \mathbb{E}\left[\frac{\hat{\sigma}^2}{n_1 n_0 / n}\right] \\ &= \frac{n}{n_1 n_0} \mathbb{E}[\hat{\sigma}^2].\end{aligned}$$

- n , n_1 , and n_0 are fixed. Now use Question 1:

$$\begin{aligned}\mathbb{E}[\widehat{V}_{\text{OLS}}] &= \frac{n}{n_1 n_0} \cdot \frac{(n_1 - 1)\sigma_1^2 + (n_0 - 1)\sigma_0^2}{n - 2} \\ &= \frac{n(n_1 - 1)\sigma_1^2 + n(n_0 - 1)\sigma_0^2}{n_1 n_0 (n - 2)}.\end{aligned}$$

Review Question 3 - Answer 3 (3)

- Subtract the true variance:

$$\begin{aligned}\text{Bias}(\hat{V}_{\text{OLS}}) &= \frac{n(n_1 - 1)\sigma_1^2 + n(n_0 - 1)\sigma_0^2}{n_1 n_0 (n - 2)} \\ &\quad - \frac{\sigma_1^2}{n_1} - \frac{\sigma_0^2}{n_0}.\end{aligned}$$

- Use the same denominator for all three terms:

$$\frac{\sigma_1^2}{n_1} = \frac{n_0(n - 2)\sigma_1^2}{n_1 n_0 (n - 2)}, \quad \frac{\sigma_0^2}{n_0} = \frac{n_1(n - 2)\sigma_0^2}{n_1 n_0 (n - 2)}.$$

- Collect the coefficients on each population variance:

$$\begin{aligned}\text{Bias}(\hat{V}_{\text{OLS}}) &= \frac{\{n(n_1 - 1) - n_0(n - 2)\}\sigma_1^2}{n_1 n_0 (n - 2)} \\ &\quad + \frac{\{n(n_0 - 1) - n_1(n - 2)\}\sigma_0^2}{n_1 n_0 (n - 2)}.\end{aligned}$$

Review Question 3 - Answer 3 (4)

- First simplify the coefficient on σ_1^2 :

$$\begin{aligned}n(n_1 - 1) - n_0(n - 2) &= nn_1 - n - nn_0 + 2n_0 \\&= n(n_1 - n_0) - n + 2n_0 \\&= n(n_1 - n_0) - (n_1 + n_0) + 2n_0 \\&= n(n_1 - n_0) - n_1 + n_0 \\&= n(n_1 - n_0) - (n_1 - n_0) \\&= (n_1 - n_0)(n - 1).\end{aligned}$$

- We used $n = n_1 + n_0$ in the third line and factored out $n_1 - n_0$ in the last line.

Review Question 3 - Answer 3 (5)

- Similarly, simplify the coefficient on σ_0^2 :

$$\begin{aligned}n(n_0 - 1) - n_1(n - 2) &= nn_0 - n - nn_1 + 2n_1 \\&= n(n_0 - n_1) - n + 2n_1 \\&= n(n_0 - n_1) - (n_1 + n_0) + 2n_1 \\&= n(n_0 - n_1) + n_1 - n_0 \\&= n(n_0 - n_1) - (n_0 - n_1) \\&= (n_0 - n_1)(n - 1) \\&= -(n_1 - n_0)(n - 1).\end{aligned}$$

- The two coefficients have the same magnitude and opposite signs.

Review Question 3 - Answer 3 (6)

- Substitute the two simplified coefficients:

$$\text{Bias}(\hat{V}_{\text{OLS}}) = \frac{(n_1 - n_0)(n - 1)\sigma_1^2}{n_1 n_0 (n - 2)} - \frac{(n_1 - n_0)(n - 1)\sigma_0^2}{n_1 n_0 (n - 2)}.$$

- Factor out the common term to obtain the desired result:

$$\text{Bias}(\hat{V}_{\text{OLS}}) = \frac{(n_1 - n_0)(n - 1)}{n_1 n_0 (n - 2)} (\sigma_1^2 - \sigma_0^2).$$

- Thus, bias depends on both the **group sizes** and the **outcome variances**.

Review Question 3 - Answer 3 (7)

- **Unbiased** if either $n_1 = n_0$ (balanced design) or $\sigma_1^2 = \sigma_0^2$ (equal variances).
- Otherwise, the sign depends on which group has the larger variance:
 - Larger group has larger variance $\rightarrow$ upward bias
 - Larger group has smaller variance $\rightarrow$ downward bias
- **Neyman variance estimator**: estimate each group's variance separately,

$$\hat{V}_{\text{Neyman}} = \frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}, \quad s_t^2 = \frac{1}{n_t - 1} \sum_{i: T_i=t} (Y_i - \bar{Y}_t)^2.$$

This is unbiased for the variance of the PATE estimator here and equals the **HC2** variance estimator in this regression.

Extra: From Imputation to Robustness (1)

- By dividing the normal equations by n_1 and n_0 , we can write the

$$\bar{Y}_1 = \hat{\alpha} + \hat{\beta} + (\hat{\gamma} + \hat{\delta})^\top \frac{1}{n_1} \sum_{i=1}^n T_i \tilde{\mathbf{X}}_i$$

$$\bar{Y}_0 = \hat{\alpha} + \hat{\gamma}^\top \frac{1}{n_0} \sum_{i=1}^n (1 - T_i) \tilde{\mathbf{X}}_i$$

where $\bar{Y}_1 = \frac{1}{n_1} \sum_{i=1}^n T_i Y_i$ and $\bar{Y}_0 = \frac{1}{n_0} \sum_{i=1}^n (1 - T_i) Y_i$

- Subtract the second line from the first and solve for $\hat{\beta}$
(difference-in-means – a correction built from the **red terms**):

$$\hat{\beta} = (\bar{Y}_1 - \bar{Y}_0) - \left\{ (\hat{\gamma} + \hat{\delta})^\top \frac{1}{n_1} \sum_{i=1}^n T_i \tilde{\mathbf{X}}_i - \hat{\gamma}^\top \frac{1}{n_0} \sum_{i=1}^n (1 - T_i) \tilde{\mathbf{X}}_i \right\}$$

Extra: From Imputation to Robustness (2)

- What are the **red terms**? Since $\tilde{\mathbf{X}}_i$ is demeaned,

$$\frac{1}{n_1} \sum_{i=1}^n T_i \tilde{\mathbf{X}}_i = \underbrace{\frac{1}{n_1} \sum_{i=1}^n T_i \mathbf{X}_i}_{\text{Avg. of treated}} - \underbrace{\frac{1}{n} \sum_{i=1}^n \mathbf{X}_i}_{\text{Avg. of everyone}}$$

$\Rightarrow$ **chance covariate imbalance** of treated (same for control)

- Imputation: $\frac{1}{n} \sum_{i=1}^n \widehat{Y}_i(1) = \hat{\alpha} + \hat{\beta}$, so

$$\underbrace{\frac{1}{n} \sum_{i=1}^n \widehat{Y}_i(1)}_{\text{Avg. imputed } Y_i(1)} = \underbrace{\bar{Y}_1}_{\text{Treated avg.}} - \underbrace{(\hat{\gamma} + \hat{\delta})^\top \frac{1}{n_1} \sum_{i=1}^n T_i \tilde{\mathbf{X}}_i}_{\text{Slope} \times \text{imbalance}}$$

Extra: From Imputation to Robustness (3): Consistency

- Look at each piece of the identity from Step 1:

$$\hat{\beta} = \underbrace{(\bar{Y}_1 - \bar{Y}_0)}_{(i)} - \left\{ \underbrace{(\hat{\gamma} + \hat{\delta})^\top}_{(iii)} \underbrace{\frac{1}{n_1} \sum_{i=1}^n T_i \tilde{\mathbf{X}}_i}_{(ii)} - \underbrace{\hat{\gamma}^\top}_{(iii)} \underbrace{\frac{1}{n_0} \sum_{i=1}^n (1 - T_i) \tilde{\mathbf{X}}_i}_{(ii)} \right\}$$

- (i) $\bar{Y}_1 - \bar{Y}_0 \xrightarrow{P} \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)] = \text{PATE}$ (Case 1)
- (ii) By LLN

$$\frac{1}{n_1} \sum_{i=1}^n T_i \tilde{\mathbf{X}}_i = \frac{1}{n_1} \sum_{i=1}^n T_i \mathbf{X}_i - \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i \xrightarrow{P} \mathbb{E}[\mathbf{X}_i] - \mathbb{E}[\mathbf{X}_i] = \mathbf{0}$$

- (iii) $\hat{\gamma} \xrightarrow{P} \gamma$, $\hat{\delta} \xrightarrow{P} \delta$: projection coefficients, **not** “true” slopes
- Therefore, $\hat{\beta} \xrightarrow{P} \text{PATE} - \{(\gamma + \delta)^\top \mathbf{0} - \gamma^\top \mathbf{0}\} = \text{PATE}$

Extra: From Imputation to Robustness (4): Efficiency (1)

- Population version of the model, with covariates centered at $\mathbb{E}[\mathbf{X}_i]$ and projection error ϵ_i :

$$T_i = 1 : Y_i = \alpha + \beta + (\gamma + \delta)^\top (\mathbf{X}_i - \mathbb{E}[\mathbf{X}_i]) + \epsilon_i$$

$$T_i = 0 : Y_i = \alpha + \gamma^\top (\mathbf{X}_i - \mathbb{E}[\mathbf{X}_i]) + \epsilon_i$$

- Same argument as (4), in the population: the regressors include $1, T_i, \tilde{\mathbf{X}}_i, T_i \tilde{\mathbf{X}}_i$, so **within each arm**

$$\mathbb{E}[\epsilon_i \mid T_i = t] = 0, \quad \text{Cov}(\epsilon_i, \mathbf{X}_i \mid T_i = t) = \mathbf{0} \quad \text{for } t \in \{0, 1\}$$

- These hold by construction of projection, **even if the model is wrong**
- Taking expectations within each arm (randomization):
 $\mathbb{E}[Y_i(1)] = \alpha + \beta$ and $\mathbb{E}[Y_i(0)] = \alpha$, so $\beta = \text{PATE}$

Extra: From Imputation to Robustness (5): Efficiency (2)

- In the sample, $\tilde{\mathbf{X}}_i$ is centered at the **sample** average, so

$$\tilde{\mathbf{X}}_i = (\mathbf{X}_i - \mathbb{E}[\mathbf{X}_i]) - \left(\frac{1}{n} \sum_{j=1}^n \mathbf{X}_j - \mathbb{E}[\mathbf{X}_i] \right)$$

where the second part is the same for all units

- Now,

$$T_i = 1: \quad Y_i - (\gamma + \delta)^\top \tilde{\mathbf{X}}_i = \alpha + \beta + \epsilon_i + (\gamma + \delta)^\top \left(\frac{1}{n} \sum_{j=1}^n \mathbf{X}_j - \mathbb{E}[\mathbf{X}_i] \right)$$

$$T_i = 0: \quad Y_i - \gamma^\top \tilde{\mathbf{X}}_i = \alpha + \epsilon_i + \gamma^\top \left(\frac{1}{n} \sum_{j=1}^n \mathbf{X}_j - \mathbb{E}[\mathbf{X}_i] \right)$$

- The shift is common to all units, but it enters with slope $\gamma + \delta$ in the treated arm and γ in the control arm

Extra: From Imputation to Robustness (6): Efficiency (3)

- By subtracting the two lines, we can write the OLS estimator as

$$\hat{\beta} - \beta \approx \underbrace{\frac{1}{n_1} \sum_{i=1}^n T_i \epsilon_i}_{\text{Treated noise}} - \underbrace{\frac{1}{n_0} \sum_{i=1}^n (1 - T_i) \epsilon_i}_{\text{Control noise}} + \underbrace{\delta^\top \left(\frac{1}{n} \sum_{j=1}^n \mathbf{X}_j - \mathbb{E}[\mathbf{X}_i] \right)}_{\text{Sample covariates} \neq \text{population}}$$

- Approximation because we ignore the fact that $\hat{\gamma}$ and $\hat{\delta}$ are estimated
- Last term: nonzero only if effects vary with covariates ($\delta \neq \mathbf{0}$)
- The three terms are uncorrelated
 - Treated and control noise come from different units (Review Q3)
 - ϵ_i is uncorrelated with $\mathbf{X}_i$ within each arm, and units are independent
- Hence,

$$\mathbb{V}[\hat{\beta}] \approx \frac{\mathbb{V}[\epsilon_i \mid T_i = 1]}{n_1} + \frac{\mathbb{V}[\epsilon_i \mid T_i = 0]}{n_0} + \frac{\mathbb{V}[\delta^\top \mathbf{X}_i]}{n}$$

Extra: From Imputation to Robustness (7): Efficiency (4)

- Difference-in-means (Review Question 3):

$$\mathbb{V}[\bar{Y}_1 - \bar{Y}_0] = \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}, \quad \sigma_t^2 = \mathbb{V}[Y_i(t)] = \mathbb{V}[Y_i \mid T_i = t]$$

- As $\text{Cov}(\epsilon_i, \mathbf{X}_i \mid T_i = t) = \mathbf{0}$:

$$\sigma_1^2 = \mathbb{V}[(\gamma + \delta)^\top \mathbf{X}_i] + \mathbb{V}[\epsilon_i \mid T_i = 1]$$

$$\sigma_0^2 = \mathbb{V}[\gamma^\top \mathbf{X}_i] + \mathbb{V}[\epsilon_i \mid T_i = 0]$$

- Subtract $\mathbb{V}[\hat{\beta}]$ from (14); the ϵ_i terms cancel:

$$\mathbb{V}[\bar{Y}_1 - \bar{Y}_0] - \mathbb{V}[\hat{\beta}] \approx \frac{\mathbb{V}[(\gamma + \delta)^\top \mathbf{X}_i]}{n_1} + \frac{\mathbb{V}[\gamma^\top \mathbf{X}_i]}{n_0} - \frac{\mathbb{V}[\delta^\top \mathbf{X}_i]}{n}$$

Extra: From Imputation to Robustness (8): Efficiency (5)

- Multiply by n , use $n_1 = np$ and $n_0 = n(1 - p)$, and expand $\mathbb{V}[\delta^\top \mathbf{X}_i] = \mathbb{V}[(\gamma + \delta)^\top \mathbf{X}_i - \gamma^\top \mathbf{X}_i]$:

$$\begin{aligned} & n\{\mathbb{V}[\bar{Y}_1 - \bar{Y}_0] - \mathbb{V}[\hat{\beta}]\} \\ & \approx \frac{\mathbb{V}[(\gamma + \delta)^\top \mathbf{X}_i]}{p} + \frac{\mathbb{V}[\gamma^\top \mathbf{X}_i]}{1 - p} \\ & \quad - \left\{ \mathbb{V}[(\gamma + \delta)^\top \mathbf{X}_i] + \mathbb{V}[\gamma^\top \mathbf{X}_i] - 2 \text{Cov}((\gamma + \delta)^\top \mathbf{X}_i, \gamma^\top \mathbf{X}_i) \right\} \\ & = \frac{1 - p}{p} \mathbb{V}[(\gamma + \delta)^\top \mathbf{X}_i] + \frac{p}{1 - p} \mathbb{V}[\gamma^\top \mathbf{X}_i] + 2 \text{Cov}((\gamma + \delta)^\top \mathbf{X}_i, \gamma^\top \mathbf{X}_i) \\ & = \mathbb{V} \left[\sqrt{\frac{1-p}{p}} (\gamma + \delta)^\top \mathbf{X}_i + \sqrt{\frac{p}{1-p}} \gamma^\top \mathbf{X}_i \right] \geq 0 \end{aligned}$$

- **Conclusion:** The fully interacted regression is never less efficient than difference-in-means, and it is more efficient if treatment effects vary with covariates ($\delta \neq \mathbf{0}$).